

# Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 2: Mechanical vs. Chemical Digestion — Breaking Down the Ugali and Sukuma Wiki

**Teacher reference only. Do not project or hand out.**

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| B  | A  | A  | C  | C  | C  | D  |

USE AFTER: PREDICT PHASE — Predict: sketch and 2-sentence prediction about what happens to ugali in the first 60 seconds in the mouth

**Q1. Before the lesson, a learner predicts: "Teeth break the ugali into small pieces and that is all that happens in the mouth." Which statement best corrects this prediction?**

- A. Chewing changes the starch in ugali into maltose
- B. Saliva contains an enzyme that also chemically changes the starch in the mouth**
- C. Saliva only makes the food wet so it is easier to swallow
- D. Chemical changes to food begin only in the stomach

**Correct answer: B.** Teeth only break food physically (mechanical digestion), but saliva contains salivary amylase which begins chemical digestion of starch in the mouth.

USE AFTER: OBSERVE PHASE — Observe: the 'Mandazi Moment' demonstration (chewing for 60 seconds and recording taste at 20, 40 and 60 seconds)

**Q2. A learner chews a plain piece of mandazi for 60 seconds without swallowing. The mandazi slowly starts to taste sweeter. What is the best explanation?**

- A. Salivary amylase is changing starch in the mandazi into maltose, a sugar**
- B. Saliva is breaking down protein in the mandazi into sugar
- C. The tongue adds sugar to the mandazi while mixing it
- D. The teeth crush the mandazi and release sugar that was already in it

**Correct answer: A.** Mandazi is starchy. Salivary amylase in saliva chemically converts starch to maltose, which tastes sweet, so the sweetening is direct evidence of chemical digestion.

USE AFTER: OBSERVE PHASE — Observe: animation of food moving from mouth to stomach, completing the Observation Table (physical vs chemical columns)

**Q3. While watching an animation of food moving from the mouth to the stomach, a learner records physical changes in the Observation Table. Which of these is an example of MECHANICAL digestion?**

- A. Stomach wall muscles churning the food**
- B. Pepsin breaking down protein in nyama choma
- C. Salivary amylase changing starch to maltose
- D. Hydrochloric acid making the stomach acidic

**Correct answer: A.** Mechanical digestion is the physical breakdown of food, such as by teeth and stomach churning. Pepsin, HCl and amylase act chemically.

USE AFTER: EXPLAIN PHASE — Explain: group sentence answering 'Why did the ugali taste sweet after 60 seconds of chewing but the nyama choma would not?'

**Q4. A learner chews ugali and it begins to taste sweet. Chewing nyama choma (grilled meat) does not make it taste sweet. What is the best reason?**

- A. Ugali is softer than nyama choma, so the teeth can break it down more
- B. Saliva only works on plant foods and not on animal foods
- C. Salivary amylase acts on starch in ugali but not on the protein in nyama choma**
- D. Nyama choma is cooked so it has no chemicals that can change

**Correct answer: C.** Enzymes are specific. Salivary amylase breaks down starch (in ugali) but cannot act on protein (in nyama choma), like a jembe designed only for one job.

USE AFTER: EXPLAIN PHASE — Explain: two-column chart of agents (salivary amylase, pepsin, HCl) and the lock-and-key / jembe analogy

**Q5. Which statement correctly describes the roles of pepsin and hydrochloric acid (HCl) in the stomach?**

- A. Pepsin digests starch and HCl digests fats
- B. Pepsin and HCl churn the food to break it into smaller pieces
- C. Pepsin breaks down protein and it works only when HCl makes the stomach very acidic**
- D. HCl is an enzyme that digests protein and pepsin makes the stomach acidic

**Correct answer: C.** Pepsin is a protein-digesting enzyme that needs the acidic conditions made by HCl. Both act chemically, while churning is done by the stomach muscles.

USE AFTER: DQB PHASE — DQB: cluster B question 'How the stomach avoids digesting itself'

**Q6. In class, the teacher asks: "If pepsin and HCl break down protein, what stops the stomach from dissolving itself?" The teacher does not answer but puts it on the Driving Question Board. Why does this question arise from what learners have just seen?**

- A. Because saliva from the mouth reaches the stomach and damages its walls
- B. Because the stomach does not use any chemical digestion
- C. Because the stomach makes HCl and pepsin, which break down protein, yet the stomach is part of a living body**
- D. Because mechanical churning in the stomach wears away its walls

**Correct answer: C.** Learners saw that the stomach produces strong acid and a protein-digesting enzyme. This raises the puzzle of why the stomach itself is not digested, which is left for a later lesson.

USE AFTER: MODEL BUILDING PHASE — Model Building: revising the Lesson 1 'food tube' sketch with [M], [C] and [M+C] labels on the mouth, oesophagus and stomach

**Q7. When revising the 'food tube' model, a learner labels the stomach with [M], [C] or [M+C]. The stomach has muscular walls that churn food and gastric glands that release pepsin and HCl. Which label is correct for the stomach?**

- A. [C] only, because the stomach only makes acid and enzymes
- B. [M] only, because the stomach only churns the food
- C. No label, because digestion starts and ends in the mouth
- D. [M+C], because it churns food and also releases pepsin and HCl**

**Correct answer: D.** Churning by the muscular walls is mechanical digestion and pepsin and HCl provide chemical digestion, so the stomach contributes to both.